

Documentation / Project Report

1. Abstract

This project presents an AI-powered Resume Screening and Application Screening System that automates the process of evaluating resumes against job descriptions. The system extracts skills from resumes, calculates ATS scores, identifies missing skills, and assists recruiters in selecting suitable candidates efficiently.

2. Introduction

Recruitment is a crucial process in organizations. Manual resume screening is time-consuming and prone to human errors. The proposed system automates resume evaluation using skill matching and ATS-based scoring techniques.

3. Problem Statement

Organizations often receive hundreds of resumes for a single job opening. Reviewing each resume manually requires significant effort and time. There is a need for an automated system that can screen resumes and rank candidates efficiently.

4. Objectives

- Automate resume screening
 - Extract candidate skills
 - Compare resumes with job descriptions
 - Calculate ATS scores
 - Identify skill gaps
 - Rank candidates
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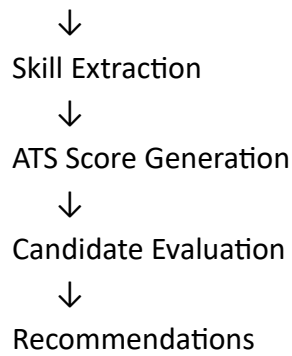
5. System Architecture

Insert:

Resume Upload



Text Extraction



6. Methodology

Step 1: Resume Upload

Users upload their resumes in DOCX format.

Step 2: Resume Parsing

Text is extracted from resumes using python-docx.

Step 3: Skill Extraction

The system identifies technical skills from the resume text.

Step 4: Job Description Matching

Resume skills are compared with job requirements.

Step 5: ATS Score Calculation

The ATS score is calculated based on matching skills.

Step 6: Skill Gap Analysis

Missing skills are identified and displayed.

7. Technologies Used

Technology	Purpose
Python	Core Programming
Streamlit	Web Application
Pandas	Data Processing
Scikit-learn	Text Similarity

Technology	Purpose
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python-docx	Resume Parsing
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GitHub	Version Control
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8. Results

Include screenshots:

- Streamlit Home Page
 - ATS Score Output
 - Matched Skills
 - Missing Skills
 - Candidate Ranking
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9. Conclusion

The developed system successfully automates the resume screening process and provides ATS-based evaluation. The system improves recruitment efficiency by reducing manual effort and identifying suitable candidates quickly.

10. Future Scope

- Support PDF resumes
- Advanced NLP models
- Multi-resume ranking
- Cloud database integration
- AI-powered career recommendations